1 · with tripod & case

Dwarf 3 Smart Telescope

WHAT'S IN THE KIT

- Dwarf 3 smart telescope
- Tripod
- Carrying case with screen cleaner, sun filter, charging cord

QUICK-START GUIDE

1. Control via the Dwarf app on a phone/tablet — it auto-finds targets.
2. Daytime: use the SUN FILTER for solar viewing (never without it).
3. Night events: the scope stacks exposures automatically for deep-sky images.
4. Images save to the app/SD — instant astronomy portfolio.

FULL LESSONS IN THIS GUIDE

PRO TIP: *See actual images taken with this scope in Littleton — ask Nick! Great hook for a family astronomy night.*

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Dwarf 3 Smart Telescope

LESSON · GRADES 3-5

Sun Spotters

LEARNING OUTCOMES

- Students will observe the sun safely with proper filtration
- Students will document changes in a natural object over time
- Students will infer rotation from position changes

MATERIALS

- Dwarf 3 telescope + SUN FILTER (non-negotiable)
- Tablet with the Dwarf app
- Sunspot sketch journal (one page per observation day)
- Clear-day backup schedule

INSTRUCTIONS

1. Safety first, loudly: the sun filter goes on before the telescope ever aims up.
2. Day 1: capture a solar image; students sketch visible sunspots and mark positions.
3. Repeat observations across 3-5 days at the same time.
4. Journals compare: have the spots moved? Appeared? Vanished?
5. Guide the inference: what would make spots parade across the face? Rotation.
6. Close with scale: how many Earths fit across one sunspot? (Often several!)

ACTIVITY

Sun Spotters — daytime astronomy where multi-day sunspot journals reveal the sun rotates and changes. Safe, jaw-dropping, and runnable during any class period with clear skies.

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Dwarf 3 Smart Telescope

LESSON · GRADES 6-8

Moon Journal Project

LEARNING OUTCOMES

- Students will capture systematic observations across a lunar cycle
- Students will model moon phases using original photographs
- Students will explain the geometry of illumination

MATERIALS

- Dwarf 3 telescope + app (evening captures — teacher or family involvement)
- Phase photo collection across the checkout window
- Phase model template (Earth-moon-sun geometry)
- Printed student photos for arrangement

INSTRUCTIONS

1. Plan the capture calendar: image the moon every clear evening of the checkout.
2. Each capture logs date, time, and what fraction appears lit.
3. Students arrange their own photos in sequence — the cycle emerges.
4. Model task: place each photo on the Earth-moon-sun geometry diagram.
5. The key question: where must the sun be for THIS photo? Repeat until it clicks.
6. Present: explain the cycle using only their own images as evidence.

ACTIVITY

Moon Journal Project — phase models built from photos the class captured themselves. Placing their own images on the geometry diagram beats any textbook's perfect drawings.

1 · with tripod & case

Dwarf 3 Smart Telescope

LESSON · GRADES 9-12

Deep-Sky Imaging Night

LEARNING OUTCOMES

- Students will queue and capture deep-sky astrophotography targets
- Students will process and interpret stacked astronomical images
- Students will research and present on observed objects

MATERIALS

- Dwarf 3 telescope + tripod + app
- Evening event logistics (astronomy night)
- Target list: nebulae, clusters, galaxies visible that season
- Research template per object: distance, type, significance

INSTRUCTIONS

1. Pre-plan targets with a season-appropriate list; students claim objects.
2. Astronomy night: students queue their targets; the scope auto-finds and stacks exposures.
3. While stacking runs, target owners share their object research with the group.
4. Next class: review captured images; discuss what stacking revealed vs. single frames.
5. Owners finalize presentations pairing their image with their research.
6. Exhibit: 'Deep Sky over Littleton' — their images, their words, publicly displayed.

ACTIVITY

> Deep-Sky Imaging Night — students queue real nebulae and galaxies, then present their own captures with research. 'This photo of the Orion Nebula was taken from Littleton, by me' is a sentence that changes kids.